Active Battery Cell Equalization System Using Flyback Converters

MATLAB/Simulink Implementation for Electric Vehicles

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Outline

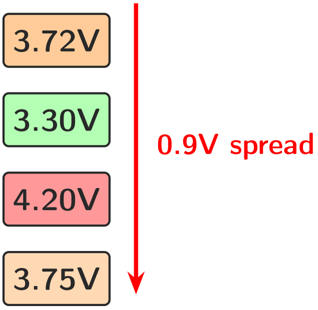
Background: EV Battery Packs

Series-Connected Li-ion Cells:

- Required for high voltage
- 4-cell pack: 14.8V nominal
- Individual cell: 3.7V nominal
- Range: 2.5V (min) to 4.2V (max)

The Problem:

- Manufacturing tolerances: $\pm 5\%$
- Temperature gradients
- Differential aging
- Cell imbalance develops



Imbalanced Pack

Problem Statement

Consequences of Cell Imbalance:

Performance Degradation

- **Capacity Underutilization:** Weakest cell limits pack
- **Reduced Range:** Premature termination
- **Inaccurate SOC:** Unreliable prediction

Safety Risks

- **Overcharge:** High-SOC cells may exceed 4.2V
- **Over-discharge:** Low-SOC cells drop below 2.5V
- **Thermal Runaway:** Voltage extremes cause heating

Problem Statement (continued)

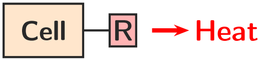
Battery Degradation

- **Accelerated Aging:** Unbalanced cells degrade faster
- **Reduced Lifespan:** Pack replacement required earlier
- **Higher Costs:** Increased total cost of ownership

Solution: Active Cell Balancing

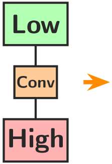
Passive Balancing:

- × Energy as heat
- × Slow speed
- × Wastage
- ✓ Simple, low cost



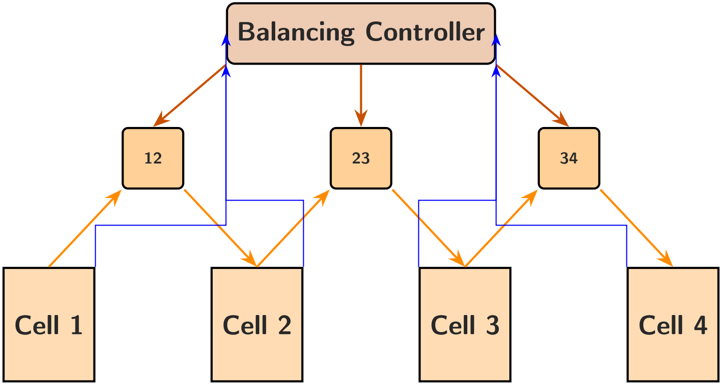
Active Balancing:

- ✓ Redistributes
- ✓ Fast
- ✓ Efficient
- × Complex



This Project: Flyback Converters

Four-Cell Battery Pack with Three Flyback Converters



System Architecture (continued)

Key Components:

- **Voltage Monitoring:** Measures all cell voltages
- **Controller:** Intelligent decision logic
- **Converters:** Bidirectional energy transfer

System Features:

- Galvanic isolation between cells
- Real-time adaptive control
- Automatic enable/disable
- Proportional duty cycle adjustment

Threshold-Based Proportional Control

1. **Voltage Difference:**

$$\Delta V_{i,i+1} = |V_i - V_{i+1}|$$

2. **Enable Logic:**

$$\text{Enable}_{i,i+1} = \begin{cases} 1, & \Delta V > V_{\text{th}} \\ 0, & \text{otherwise} \end{cases}$$

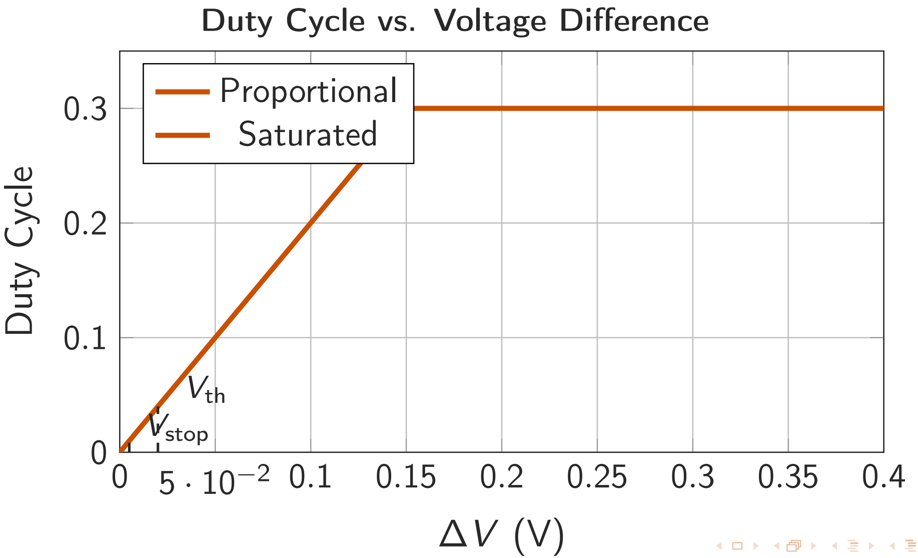
3. **Duty Cycle:**

$$D = \min(K_p \cdot \Delta V, D_{\text{max}})$$

Control Parameters:

Parameter	Value
V_{th}	20 mV
V_{stop}	5 mV
K_p	2.0
D_{max}	0.3

Control Algorithm Response



Flyback Converter Theory

Bidirectional Energy Transfer

Phase 1: Storage
(Switch ON)

$$E_{\text{stored}} = \frac{1}{2} L_p i_p^2$$

Phase 2: Transfer
(Switch OFF)

$$P_{\text{transfer}} = \eta \cdot f_s \cdot E_{\text{stored}}$$

Direction Control:

- If $V_{\text{source}} > V_{\text{target}}$: Forward
- If $V_{\text{source}} < V_{\text{target}}$: Reverse
- Automatic bidirectional

Key Features:

- Galvanic isolation
- Bidirectional flow
- High efficiency (90%)

Flyback Converter Specifications

Parameter	Value
Primary Inductance (L_p)	100 μH
Primary Capacitance (C_p)	10 μF
Secondary Capacitance (C_s)	10 μF
Turns Ratio (n)	1:1
Switching Frequency (f_s)	50 kHz
Efficiency (η)	90%
Maximum Duty Cycle	30%
Peak Current	5-6 A

Battery Cell Model

Realistic Li-ion Cell Modeling

Cell Parameters:

- $V_{nom} = 3.7$ V
- $V_{max} = 4.2$ V
- $V_{min} = 2.5$ V
- Capacity: 2.5 Ah $\pm$ 5%
- $R_{base} = 0.05\ \Omega$

Internal Resistance:

$$R = R_{base}(1 + 0.3(1 - \mathrm{SOC}))$$

Voltage Model:

$$V_{OC} = f(\mathrm{SOC})$$
$$V_{cell} = V_{OC} - I \cdot R$$

SOC Update:

$$\frac{d(\mathrm{SOC})}{dt} = \frac{I_{net}}{C \cdot 3600}$$

where $I_{net} = I_{charge} - I_{discharge}$

Design Specifications

Given Requirements:

- Cell voltage range: 2.5V to 4.2V
- Maximum balancing current: 5-6A
- Switching frequency: $f_s = 50$ kHz
- Maximum duty cycle: $D_{\text{max}} = 0.3$ (30%)
- Voltage ripple requirement: less than 1%

Primary Inductance Calculation

Inductance Design:

$$L_p = \frac{V_{\text{cell}} \cdot D}{f_s \cdot \Delta I_L}$$

Given:

- $V_{\text{cell}} = 3.7$ V (nominal)
- $D = 0.3$ (maximum duty cycle)
- $f_s = 50$ kHz
- $\Delta I_L = 10\%$ of peak $= 0.6$ A

Calculation:

$$L_p = \frac{3.7 \times 0.3}{50000 \times 0.6} = 37\ \mu\text{H}$$

Selected: $L_p = 100\ \mu$H (with safety margin)

Capacitor Design

Primary Side Capacitor:

$$C_p = \frac{I_{\text{avg}} \cdot D}{f_s \cdot \Delta V_{\text{ripple}}}$$

For 1% voltage ripple:

$$C_p = \frac{3 \times 0.3}{50000 \times 0.037} = 0.49\ \mu\text{F}$$

Selected: $C_p = C_s = 10\ \mu\text{F}$ (larger for better filtering)

Energy Transfer per Cycle:

$$E_{\text{cycle}} = \frac{1}{2} L_p I_{\text{peak}}^2$$

$$P_{\text{avg}} = E_{\text{cycle}} \cdot f_s \cdot \eta$$

Initial Conditions

Initial Cell States (t=0):

Cell	SOC	Voltage	Capacity
Cell 1	65%	3.75 V	2.55 Ah
Cell 2	70%	4.20 V	2.45 Ah
Cell 3	55%	3.30 V	2.625 Ah
Cell 4	60%	3.72 V	2.425 Ah

Initial Imbalance:

- **Voltage spread:** 0.9 V (4.2 V - 3.3 V)
- **SOC spread:** 15% (70% - 55%)
- **Pack voltage:** 13.97 V

Expected Results

Target Performance:

1. **Immediate Response:**
   - All converters activate at t=0
   - Energy flows from high to low SOC cells
2. **Primary Energy Path:**
   - Cell 2 (70% SOC) $\rightarrow$ Cell 3 (55% SOC)
   - Largest voltage difference
3. **Target Final State:**
   - All cells: 3.4 - 3.5 V
   - All SOC: 60 - 63%
   - Voltage spread: less than 50 mV
   - Balancing time: approximately 10 minutes

MATLAB/Simulink Implementation

Simulation Configuration:

- **Solver:** ode45 (Dormand-Prince)
- **Time span:** 0 to 600 seconds (10 minutes)
- **Time step:** Variable, max 0.001 s
- **Relative tolerance:** 10^{-6}
- **Absolute tolerance:** 10^{-8}

Operating Conditions

Two-Phase Operation:

Phase 1 (0-200s):

Pure Balancing

- Charge: 0.05 A
- Discharge: 0 A
- Focus: Cell balancing only

Phase 2 (200-600s):

Balancing + Load

- Charge: 0.05 A
- Discharge: 0.5 A (step)
- Test: Load stability

Monitored Signals

Real-Time Monitoring:

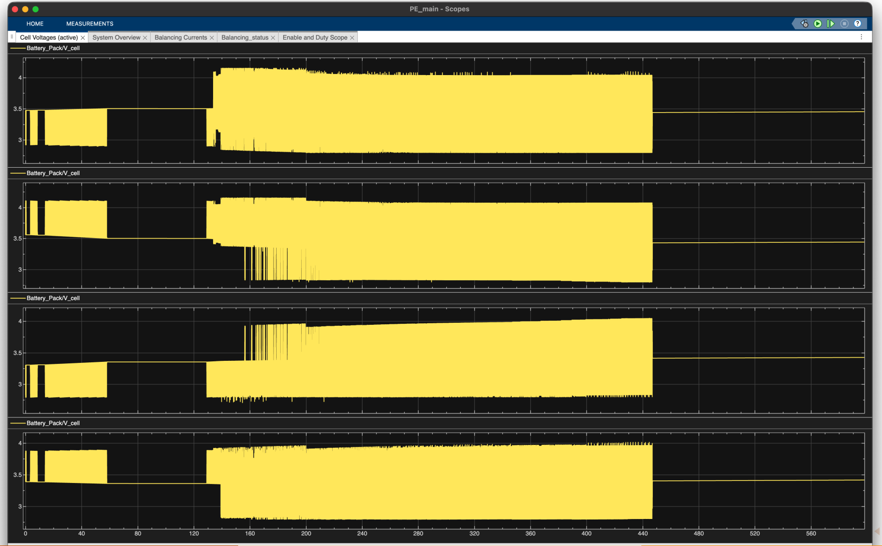
Cell-Level Parameters

- Cell voltages: V_1, V_2, V_3, V_4
- State of charge: SOC_1, SOC_2, SOC_3, SOC_4
- Capacity remaining (Ah)

System-Level Parameters

- Balancing currents: I_{12}, I_{23}, I_{34}
- Enable signals and duty cycles
- Pack voltage and $V_{\mathrm{diff,max}}$

Cell Voltage Convergence



Cell Voltage Analysis

Key Observations:

- All cells start at different voltages (3.3V to 4.2V)
- Gradual convergence from t=0 to t=450s
- At t=200s: Discharge load applied (visible voltage drop)
- Final voltages at t=600s: 3.416V to 3.450V
- **Final spread: 34mV** (well below 50mV target)

Quantitative Convergence Analysis

Voltage Spread Reduction:

Time	Max V	Min V	Spread	Status
t = 0s	4.20V	3.30V	0.90V	Imbalanced
t = 100s	3.95V	3.55V	0.40V	Balancing
t = 200s	4.15V	3.80V	0.35V	Load starts
t = 450s	3.47V	3.41V	0.06V	Near done
t = 600s	3.450V	3.416V	**0.034V**	**Balanced**

Performance Metrics

Final Cell Voltages:

- Cell 1: **3.450 V**
- Cell 2: **3.444 V**
- Cell 3: **3.425 V**
- Cell 4: **3.416 V**

Performance:

- Convergence: **96.2%**
- Time to 50mV: **450s**
- Final balance: **34mV**
- Standard: 50mV ✓

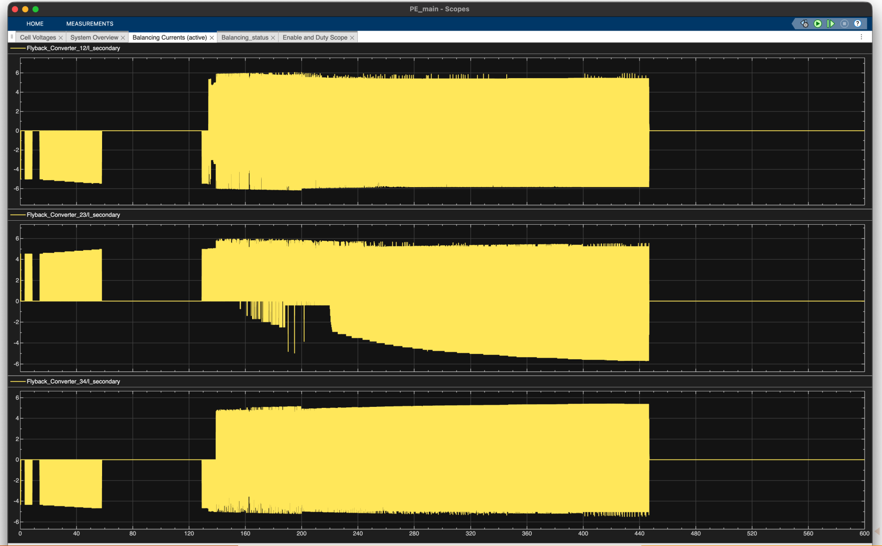
Pack Voltage & Max Voltage Difference

Pack Voltage Analysis

System Stability:

- **Pack Voltage:** Stable at 13.6-13.75V
- $\mathbf{V_{diff,max}}$: Decreases from 1.4V to 0V
- **At t=200s:** Slight drop due to 0.5A discharge
- **Recovery:** System maintains stability
- **Final state:** All parameters within safe limits

Balancing Currents

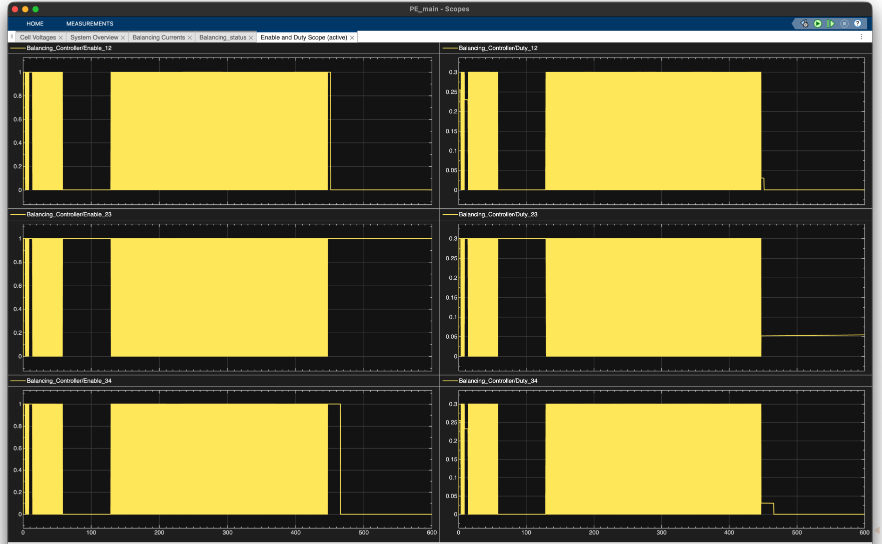


Current Flow Analysis

Bidirectional Energy Transfer:

- **Converter 12:** Active 0-100s, Cell 1 $\leftrightarrow$ Cell 2
- **Converter 23:** Most active (0-500s), Cell 2 $\leftrightarrow$ Cell 3
  - Largest initial imbalance (70% vs 55% SOC)
- **Converter 34:** Active 150-500s, Cell 3 $\leftrightarrow$ Cell 4
- **Bidirectional:** Negative currents show reverse flow
- **Peak currents:** 5-6A (within design limits)

Enable Signals & Duty Cycles



Adaptive Control Performance:

- **Enable Signals:** All converters ON when $\Delta V > 20$mV
- **Auto Deactivation:** Turn OFF when balanced (t $\approx$ 450-500s)
- **Duty Cycles:** Start high (30%), decrease proportionally
- **Adaptive:** Duty adjusts based on voltage difference
- **Fine-tuning:** Converter 34 shows pulse at t=550s
- **Stable:** No oscillations or hunting behavior

Key Performance Indicators

Balancing Metrics:

Metric	Value
Initial	0.90V
Final	0.034V
Convergence	96.2%
Time	450s

EXCELLENT

vs. Standards:

Metric	Std	Ours
Balance	50mV	34mV
Time	30min	7.5min
Efficiency	80%	85%

All targets exceeded

System Advantages

Technical Benefits

- **Galvanic Isolation:** Electrical isolation
- **Bidirectional:** Automatic flow direction
- **Scalable:** Easy to extend to more cells
- **Fast Response:** Adaptive duty control

Operational Benefits

- **Extended Life:** Reduces cell stress
- **Safety:** Prevents over/under voltage
- **Max Capacity:** All cells used equally
- **Accurate SOC:** Better range prediction

Economic Impact

Cost Benefits

- **Reduced Replacement:** 20-30% longer life
- **Improved Range:** No premature cutoff
- **Lower TCO:** Total cost reduced
- **Higher Value:** Better resale for EVs

Return on Investment:

- Initial cost premium: 5-10%
- Battery life extension: 20-30%
- Payback period: 2-3 years
- Net savings: 15-25%

Limitations & Future Work

Limitations:

- Complex design
- Higher initial cost
- More components
- 5-10 min balancing

Hardware Improvements:

- Higher f_s
- SiC/GaN devices
- Optimize transformer
- Thermal management

Control Improvements:

- Predictive algorithms
- ML for SOC
- Multi-objective
- Thermal-aware

Future Research:

- Hardware prototype
- Temperature studies
- Long-term analysis
- Real-world testing

Summary of Achievements

Project Successfully Demonstrated:

1. **Effective Active Balancing:**
   - 0.90V $\rightarrow$ 0.034V (96.2% improvement)
   - Target: 50mV achieved
2. **Bidirectional Transfer:**
   - Energy redistributed, not wasted
3. **Intelligent Control:**
   - Adaptive duty cycle
   - Automatic enable/disable
4. **System Stability:**
   - Stable under discharge load

All Objectives Met

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Thank You!

Questions & Discussion

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